

AI-BASED APPLICATIONS · FINAL PROJECT

Leuphana University Lüneburg

RAG RANGERS

AI PDF Chat Application

An AI-Powered RAG System for Sustainability Report Analysis

Eugene

Max

Elias

Finn



OVERVIEW

Agenda



01 Project Goals



02 Development Journey



03 Technology Stack



04 System Architecture



05 PDF Processing Pipeline



06 RAG Answer Pipeline



07 Frontend + User Interface



08 Testing & Benchmarking



09 Challenges



10 Meet the Rangers



11 Final Results & Repos



12 Live Demo



THE TEAM

Meet the Rangers



Eugene

Scrum Master

Presentation



Max

RAG Pipeline

PDF Processing



Elias

Streamlit UI

Main Structure



Finn

Benchmarking

Code Cleanup



OUR MISSION

Project Goals



Upload Reports

Upload sustainability reports as PDFs.



Ask Questions

Ask questions about the uploaded document.



Retrieve Context

Retrieve relevant context using RAG.



Grounded Answers

Generate grounded answers with SAIA / Gemma.



Export JSON

Export structured results as JSON.



HOW WE GOT HERE

Development Journey

1



Project
Planning

2



SAIA API
Integration

3



PDF
Processing

4



RAG Pipeline

5



Frontend
Development

6



Benchmarking
& Testing

7



Final
Application



BUILT WITH

Technology Stack

CORE TECHNOLOGIES



Python



Streamlit



ChromaDB



GWDG SAIA



Gemma

DEVELOPMENT TOOLS



GitHub



Trello



ChatGPT



Gemini

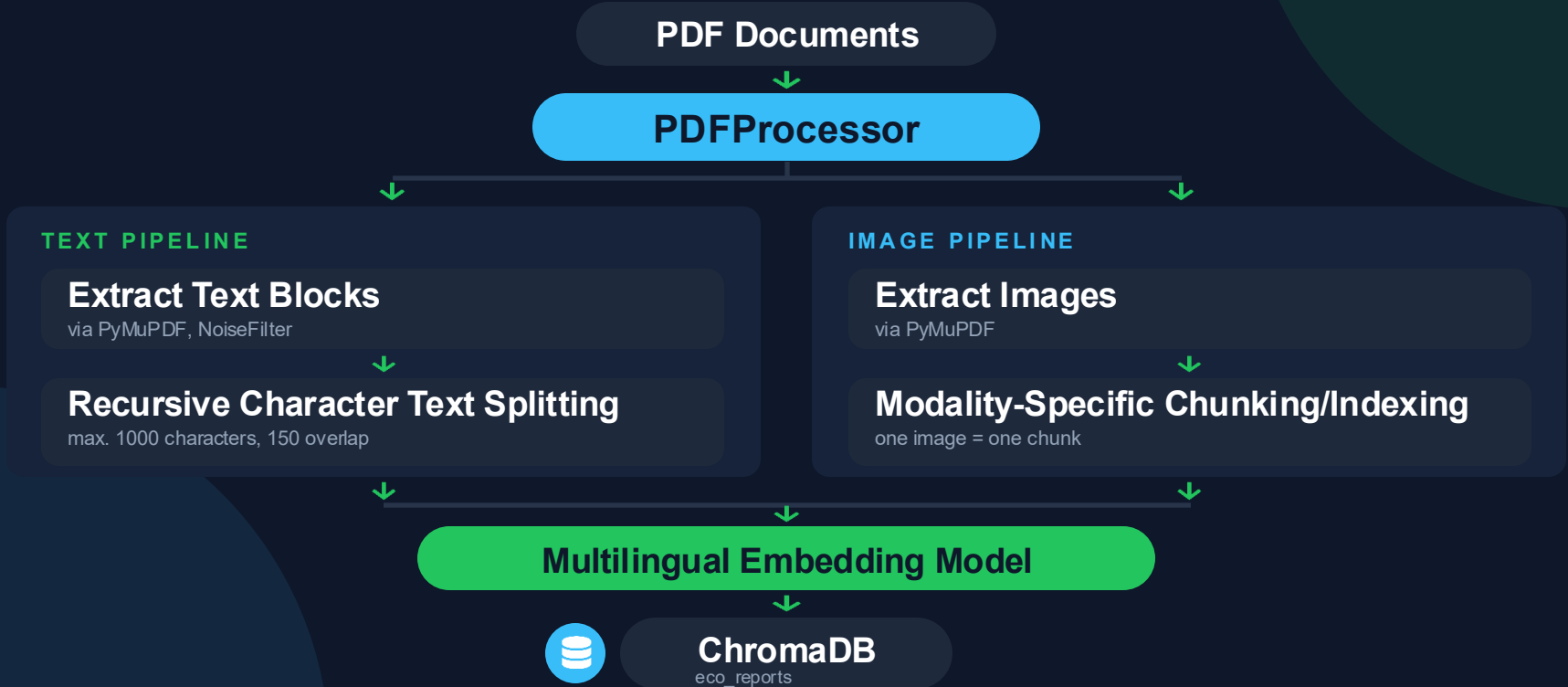
ChatGPT and Gemini were used as AI-assisted development tools for coding support, debugging, documentation and presentation design. The overall system architecture, implementation, testing and integration were completed by the Sustainable RAG Rangers team.



MAX'S PART

PDF Processing Pipeline

How sustainability reports are prepared for retrieval





MAX'S PART

Embedding Model

Why use Multilingual-e5-base?

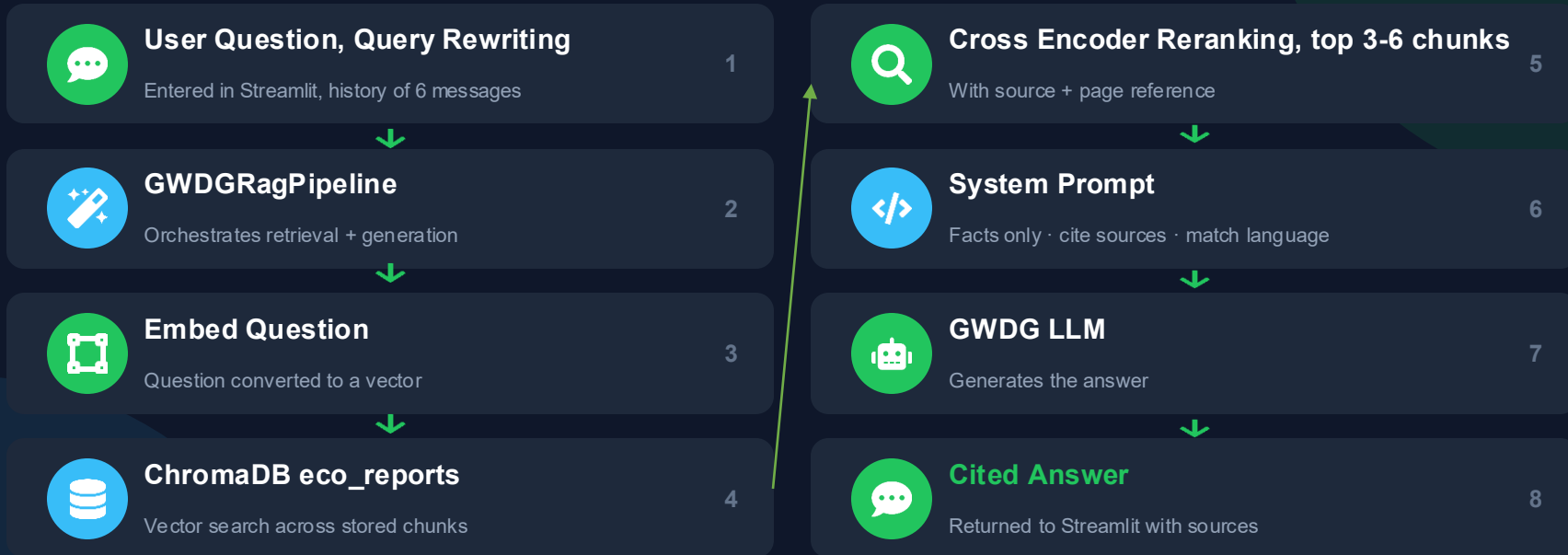
- **Before: BAAI/bge-m3 – too slow for CPU-only inference**
- **Switch: 260 seconds → 75s embedding time per 15 MB report**
- **Trade-Off: smaller model, still solid DE/EN retrieval quality**



MAX'S PART

RAG Answer Pipeline

How a user question becomes a grounded answer



Retrieval and generation work together: only cited, grounded content is returned to the user.



MAX'S PART

System Prompt

```
system_prompt = (  
    "You are a highly qualified academic assistant and subject matter expert for analyzing corporate and sustainability reports. "  
    "Your task is to answer user questions precisely, objectively, and factually.\n\n"  
    "STRICTLY adhere to the following rules:\n"  
    "1. FACT-BASED ONLY: Answer the question SOLELY based on the provided PDF context below. Do not use any external knowledge.\n"  
    "2. ANTI-HALLUCINATION: If the provided context does not contain sufficient information to answer the question, do not guess. "  
    "Reply exactly with: 'The provided document does not contain information on this topic.' (or the equivalent in the user's language).\n"  
    "3. STYLE & TONE: Formulate your answers in a clean, professional, and objective academic tone. Use complete sentences.\n"  
    "4. MULTILINGUAL (CRITICAL): You MUST answer in the EXACT SAME LANGUAGE as the user's question. "  
    "If the user asks in English, write your entire response in English. If the user asks in German, write your entire response in German. "  
    "Match the language perfectly.\n"  
    f"{citation_rule}"  
)
```



Benchmarking LM's

ArcelorMittal_SA-SASB_Index

Q: Which specific international frameworks and standards does ArcelorMittal use as a basis for its 2021 Reporting Index?

A: The index is based on the principles of the International Integrated Reporting Framework (IIRC), the Global Reporting initiative (GRI) Sustainability REporting Standards, and the Sustainability Accounting Standards Board (SASB) standards

Amazoncom_Inc-2021_Sustainability_Report

Q: What is Amazons current target year to power its entire operations with 100% renewable energy, and how many years ahead of its original target is this?

A: Amazon is on the path of powering its operations with 100% renewable energy by 2025 which is five years ahead of its original target of 2030.

Q: How much did Amazon invest and what does the Upskilling 2025 pledge consist of?

A: Amazon has invested \$1.2 Billion to upskill 300,000 of their own employees by 2025

Q: What is Amazons global Men/Women ratio for Senior Leaders in the year 2020?

A: The Role "Senior Leaders" in the year 2020 has had a Men/Women Split of 77.9% for Men and 21.2% for Women globally. (Page 99.)

Red= Questions regarding graphs or visualisations

Green= Questions regarding tables

Models Tested:

1. gemma-4-31b-it
2. meta-llama-3.1-8b-instruct

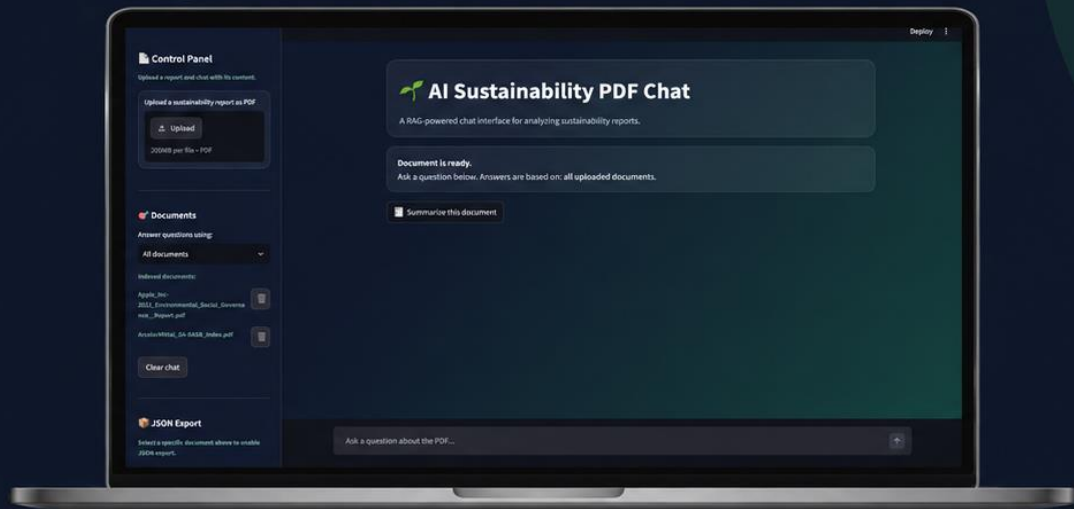
Results:

Gemma wins with response time of 7 seconds on average.



ELIAS'S PART

Frontend Implementation + User Interface



● Sidebar

● PDF Upload

● Chat Interface

● AI Response

● JSON Export



Streamlit UI

- PDF Upload
- Chat Input
- AI Response Display
- Status Indicators
- JSON Export Button



Backend Connection

- Calls PDF Processing
- Triggers RAG Pipeline
- Sends context to SAIA API
- Enables JSON Export

An early frontend prototype was built with React + Vite. The team later migrated to Streamlit because it integrated better with the Python-based backend and RAG modules.



REFLECTION

Lessons Learned



Teamwork & Communication

Clear ownership of modules kept four parallel workstreams aligned.



Iterate Early

Committing to one stack (Streamlit) early would have saved rework.



Ground Truth Matters

Strict RAG grounding avoided hallucinated answers.



WHAT WE SHIPPED

Project Results



Working PDF Upload



Working Chat



Working RAG



Working Streamlit UI



JSON Export



Benchmarking

ai-pdf-chat-app main

Project

ai-pdf-chat-app ~/PycharmProjects/ai-pdf-chat-app

.claude

.venv

app

json_export.py

main.py

pdf_processing.py

rag_pipeline.py

chroma_db

docs

extracted_images

frontend (Plan B)

pdfs

.env

.gitignore

check_models.py

README.md

requirements.txt

External Libraries

Scratches and Consoles

main.py

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import html

import re

import time

from io import BytesIO

from pathlib import Path

import streamlit as st

from openai import APIConnectionError, APIStatusError, RateLimitError

from json_export import create_json_export, convert_json_export_to_string, create_chat_history_export

from pdf_processing import PDF_DIR, PDFProcessor

from rag_pipeline import get_pipeline, is_summary_question

Vorgefertigte Frage für den Summary-Button: enthält "summar", damit is_summary_question()

sie zuverlässig erkennt, unabhängig von der Sprache des restlichen Chats.

SUMMARY_BUTTON_QUESTION = "Please provide a summary of this document."

Sets the browser tab title, app icon, and page layout for the Streamlit UI.

st.set_page_config(

page_title="AI PDF Chat App",

page_icon="🗨️",

layout="wide",

)

Defines the visual design of the app, including the dark background, cards, sidebar, and chat styling.

st.markdown(

body: ""

<style>

.stApp {

background: linear-gradient(135deg, #0f172a 0%, #111827 45%, #064e3b 100%);

6 18 3 90

10:1 LF UTF-8 4 spaces Types: Built-in Python 3.14 (ai-pdf-chat-app)



ai-pdf-chat-app

Public

Watch 0

main

1 Branch 0 Tags

Go to file

T

Add file

Code

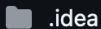


ManiDevL

Improve Streamlit chat UI and add source page previews

a0eeefb · 16 hours ago

15 Commits



.idea

feat: Basic RAG Pipeline erfolgreich mit GWDG gemma 3 v...

2 months ago



app

Improve Streamlit chat UI and add source page previews

16 hours ago



docs

Create project_notes.md

2 months ago



frontend (Plan B)

Plan B

2 weeks ago



.gitignore

Initial commit

2 months ago



README.md

Initial commit

2 months ago



check_models.py

feat: Basic RAG Pipeline erfolgreich mit GWDG gemma 3 v...

2 months ago



requirements.txt

Added chunking structure and modified pipeline accordingly

4 days ago

README



Product Backlog 0 ...

+ Eine Karte hinzufügen

Sprint Backlog 10 ...

✓ Insert App Screenshot into Presentation

✓ Insert GitHub Link / QR Code

✓ Insert Trello Link / QR Code

✓ Prepare Demo Questions

✓ Final Rehearsal

Eugen: Final Presentation
 JSON Export

Max: Finish RAG Pipeline
 Finish PDF Processing

Elias: Improve Streamlit UI
 Refactor main.py

Finn: Benchmark AI Models

✓ Final Code Cleanup

+ Eine Karte hinzufügen

In Progress 6 ...

JSON Export final polish

Final Demo Preparation

GitHub & Trello QR Codes

✓ Implement JSON Validation & Export: Write a function that parses the LLM's text response, validates that it is proper JSON, and prepares it as a .json file for download

✓ Build JSON Download Button: Add a button in the UI that allows the user to download the generated .json file from the extraction step

✓ Benchmark questions to find out which AI-model is best for our project

+ Eine Karte hinzufügen

To Review 3 ...

Final App Test Check full workflow before presentation

Presentation Review Check slides, screenshots and speaking order

Demo Review Test demo questions and fallback plan

+ Eine Karte hinzufügen

Done 9 ...

✓ Scrum Board Setup Product, Sprint, In Progress, Review and Done columns created

✓ GitHub Repository Setup Basic project structure created

✓ SAIA Integration API keys configured and connection tested

✓ PDF Processing PDF upload, text extraction and chunking completed

✓ RAG Pipeline Vector DB, retriever and final RAG chain completed

✓ Streamlit Frontend UI, PDF upload and chat interface implemented

✓ JSON Export Structured export and download function implemented

✓ Benchmarking & AI Testing Model behavior and answer quality evaluated

+ Eine Karte hinzufügen

+ Eine weitere Liste hinzufügen



FIND US

GitHub & Trello



Repository

github.com/team/ai-pdf-chat

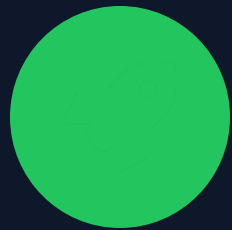
Repository placeholder — update before submission



Project Board

<https://trello.com/b/ddu5MTsK/chat-with-a-pdf>

Repository placeholder — update before submission



Thank You

Questions?

Eugene

Max

Elias

Finn